

Lesson 27 — Connecting a Trained Model to a Build

Matches LEGO Teacher Portal lesson: "Dance Trainer".

Objective (I can...): I can use my trained classifier's output as the input to a conditional that drives my LEGO build. **CS concept:** Model output as program input; classification feeding selection · **Builds on:** Lesson 26 (training) and the conditionals unit. *Privacy: on-device in the browser, no login, images not saved.*

AI Connection (2 min): The classifier's label and confidence become the condition in an IF/THEN — the same selection logic from Unit 3, now driven by AI. *First-principles question:* "Should the build still act if the model is only 55% sure? Who decides the cutoff?"

Materials: Device per pair (Teachable Machine + Coding Canvas), LEGO CS & AI kit, category cards/objects, notebooks. **Model reuse:** AI responder model from Lesson 26 — mostly code.

60-minute flow - Warm-up (8): "Label then act" — pairs decide, for three labels, what the build should do for each, and a confidence cutoff below which it does nothing. - **Concept + demo (10):** Demo reading the classifier's label and confidence, then an if/then: if label is "A" and confidence above threshold, then the build does X; else it waits. Wrap the reaction in a function from Unit 5. - **Build/Code (30):** Pairs connect their trained model so a recognized category drives the model (classify → if label/confidence → function that acts). They log the confidence threshold they chose and why. *Swap at 15 min.* - **Challenge (7):** Add a second category that calls a different reaction function. - **Share + pack-up (5):** Pairs demo the classifier driving the build and explain their confidence cutoff. Model stays built.

Pair roles: Driver codes the if/then and functions, Navigator presents items to the camera and reads label/confidence. *Swap at 15 min. (Trio: third logs thresholds.)*

Differentiation: *Support* — one label-to-action link provided; students add the second. *Extension / fast-finisher* — three labels, each a different function, with a confidence gate. *Reteach* — narrate "label = A, confidence high → true → act" while testing.

Exit check: Pair's build reacts correctly and only above a stated confidence threshold based on the classifier's output. **At-home (optional):** Describe a product that acts on an AI's guess; what should it do when the AI is unsure? ~10 min.